

Bank of Leeds Strategic Investment Proposal

Yorkshire Renewable Energy Renaissance 2050

A £6 Billion Programme to Build a Regional Energy Economy

Prepared for:

Bank of Leeds – Energy, Infrastructure & Industrial Investment Division

Investment Classification:

Renewable Energy Infrastructure / Industrial Strategy / Regional Development

Investment Horizon:

25–40 years

Executive Summary

The Bank of Leeds proposes the creation of the:

Yorkshire Renewable Energy Investment Fund

A long-term capital programme to transform Yorkshire into one of Britain's leading renewable energy regions.

The strategy would combine Yorkshire's:

- industrial land,
- former mining infrastructure,
- agricultural resources,
- coastline,
- rivers,
- geological assets,
- existing energy networks,

to create a diversified renewable energy system based on:

1. **Solar energy**
2. **Hydroelectric power**
3. **Geothermal energy**
4. **Biomass and biogas**
5. **Energy storage**
6. **Green hydrogen**

The Yorkshire and Humber region has previously been assessed as having significant renewable energy potential, with studies identifying thousands of megawatts of possible renewable capacity across technologies including biomass, solar, hydro and renewable heat.

Investment Summary

Yorkshire Renewable Energy Fund

Total Investment:

£6 Billion

Over 15 years

Sector	Investment
Solar farms & agrivoltaics	£1.5bn
Geothermal energy	£750m
Hydro & pumped storage	£500m
Biomass & biogas	£1bn
Hydrogen economy	£750m
Battery storage	£750m
Grid infrastructure	£750m

Strategic Vision

"Yorkshire Energy Valley"

A regional energy system where:

- farms produce energy,
 - mines become energy assets,
 - industrial sites become clean energy hubs,
 - ports export energy technology,
 - cities consume locally produced power.
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Pillar 1

Yorkshire Solar Renaissance

Investment:

£1.5 Billion

Opportunity

Yorkshire has extensive:

- agricultural land,
- brownfield industrial sites,
- former mining areas,
- warehouse roofs.

Solar deployment is already a major area of renewable development activity in Yorkshire and Humber, with large-scale solar projects progressing through the planning system.

Proposal

Create:

Yorkshire Solar Corporation

Develop:

100 Solar Farms

Average:

20MW each

Total capacity:

2GW

Locations:

- Selby
- Goole
- Doncaster
- East Yorkshire

- former coalfield areas
-

Agrivoltaics Programme

Investment:

£300m

Combine:

- sheep grazing,
 - crops,
 - solar generation.
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Financial Model

Installed:

2GW

Annual generation:

≈1.9 TWh

Electricity value:

£150m–£250m/year

Economic benefits:

- land income for farmers,
 - rural jobs,
 - cheaper industrial electricity.
-

Pillar 2

Yorkshire Geothermal Energy Programme

Investment:

£750 million

Strategic Opportunity

Yorkshire's industrial history has created unique geothermal opportunities.

Former mining areas contain:

- mine water systems,
- deep geological formations,
- abandoned shafts.

The UK has been investigating mine-water geothermal systems as a potential source of low-carbon heat, particularly in former coalfield areas.

Proposal

Yorkshire Geothermal Corporation

Projects:

Sheffield Mine Heat Network

Investment:

£150m

Serve:

- universities,
 - hospitals,
 - housing.
-

Barnsley Coalfield Geothermal

Investment:

£200m

Doncaster Deep Geothermal

Investment:

£250m

York Basin Geothermal

Investment:

£150m

Output

Potential:

300MW thermal capacity

Heat:

- homes,
 - industry,
 - greenhouses.
-

Economic Impact

Potential:

50,000 homes heated

Annual savings:

£20m–£50m

Pillar 3

Yorkshire Hydro & Pumped Storage

Investment:

£500 million

Hydro has a smaller role than solar or biomass in Yorkshire because of geography, but regional assessments identify hydro as a possible contributor, particularly through smaller schemes.

Proposal

Small Hydro Network

Locations:

- Pennine valleys
 - Yorkshire Dales
 - canal infrastructure
-

Projects:

50 small hydro installations

Average:

1MW

Capacity:

50MW

Investment:

£150m

Pumped Storage

Use:

- abandoned mines,
- reservoirs,
- quarries.

Investment:

£350m

Purpose:

- grid balancing,
 - renewable storage.
-

Pillar 4

Yorkshire Biomass & Biogas Network

Investment:

£1 Billion

Yorkshire already has major biomass energy infrastructure, including the large-scale biomass operation at Drax in North Yorkshire. Regional studies have identified biomass as one of the major renewable resource opportunities for Yorkshire and Humber, including agricultural residues, waste streams and energy crops.

Proposal

Create:

Yorkshire Circular Energy Company

Sources:

- farm waste,
 - food waste,
 - sewage,
 - agricultural residues.
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Projects:

50 Anaerobic Digesters

Investment:

£500m

Output:

200MW equivalent

Farm Energy Programme

Investment:

£250m

Includes:

- methane capture,
 - CHP systems,
 - farm digesters.
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Waste-to-Energy

Investment:

£250m

Financial Model

Annual revenue:

£200m–£300m

Benefits:

- reduces waste costs,
 - creates rural income,
 - improves energy security.
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Pillar 5

Yorkshire Green Hydrogen Economy

Investment:

£750 million

Vision

Use renewable electricity to produce hydrogen for:

- steel,
 - transport,
 - industry.
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Locations:

- Sheffield
 - Rotherham
 - Hull
 - Teesside corridor
-

Projects:

Electrolysis Plants

Capacity:

300MW

Investment:

£500m

Hydrogen Storage

Investment:

£250m

Potential users:

- steelmaking,
 - freight,
 - shipping,
 - chemicals.
-

Pillar 6

Yorkshire Energy Storage Network

Investment:

£750 million

Renewable energy requires storage.

Develop:

Battery Parks

Locations:

- Doncaster
 - Sheffield
 - Leeds
 - Hull
-

Capacity:

2GWh

Investment:

£500m

Former Mine Storage

Investment:

£250m

Ten-Year Financial Model

Year	Revenue	EBITDA
Year 1	£50m	£10m
Year 5	£400m	£120m
Year 10	£900m	£300m
Year 15	£1.3bn	£500m

Employment Impact

Construction

40,000 job-years

Permanent Jobs

Direct:

8,000

Indirect:

20,000+

GDP Impact

Estimated Yorkshire economic impact:

£2–3 billion annually

Through:

- energy production,
 - manufacturing,
 - engineering,
 - exports,
 - lower energy costs.
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Industrial Benefits

Sheffield Manufacturing

Cheaper electricity supports:

- steel,
 - advanced manufacturing,
 - engineering.
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Yorkshire Agriculture

Farmers gain:

- energy income,
 - waste revenue,
 - diversification.
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Coastal Economy

Supports:

- offshore wind supply chain,
 - hydrogen exports,
 - ports.
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Bank of Leeds Investment Structure

Yorkshire Renewable Infrastructure Fund

Capital:

£6 billion

Funding:

Source	Amount
Bank of Leeds	£1.5bn
Pension funds	£2bn
Government	£1bn
Private investors	£1.5bn

Recommended Priority Projects

Phase One (Years 1–5)

Investment:

£2bn

Focus:

1. Solar farms
 2. Mine geothermal
 3. Anaerobic digestion
 4. Battery storage
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Phase Two (Years 5–10)

Investment:

£2bn

Focus:

1. Hydrogen
 2. Industrial energy hubs
 3. Pumped storage
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Phase Three (Years 10–20)

Investment:

£2bn

Focus:

1. Export energy technology
 2. Large-scale storage
 3. Advanced geothermal
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Strategic Recommendation

The Bank of Leeds should establish:

Yorkshire Renewable Energy Corporation

A regional energy company owning:

- generation assets,
 - storage,
 - heat networks,
 - hydrogen infrastructure.
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Final Investment Thesis

Yorkshire's industrial revolution was powered by:

- coal,
- water,
- geology.

Its next industrial revolution can be powered by:

- sunlight,
- underground heat,
- agricultural resources,
- industrial innovation.

The Bank of Leeds opportunity is to create:

"Yorkshire Energy Valley — Britain's first integrated regional renewable energy economy."

A £6 billion investment programme could create a new strategic asset base, reduce energy costs for Yorkshire businesses, create thousands of skilled jobs and position the region as a leader in the UK's transition to a low-carbon industrial economy.